CCC–C–443c
May 1, 1967

SUPERSEDING
Fed. Spec. CCC–C–443b
March 28, 1963 (See 6.3)

FEDERAL SPECIFICATION

CLOTH, DUCK, COTTON: (SINGLE AND PLIED FILLING YARNS, FLAT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for unbleached flat duck.

1.2 Classification.

1.2.1 Types. The flat duck cotton cloth shall be of the following types as specified (see 6.2):

Type I —Single filling.
Type II—Double filling (plied).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

V–T–276 —Thread, Cotton.
CCC–T–191—Textile Test Methods.
PPP–P–51 —Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed. Std. No. 751—Stitches, Seams, and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Yarns. Yarn shall be made from cotton which has been cleaned, carded, drawn, and spun into single yarn for both warp and filling for type I and single yarn for warp, and plied yarns for filling for type II.

3.2 Physical requirements. The cloth covered by this specification shall be as specified (see 6.2) and shall conform to the requirements specified in table I when tested as specified in 4.4.

TABLE I Physical requirements

Type	Weight per square yard minimum	Yarn ply minimum		Yarns per inch minimum		Breaking strength [2] pounds minimum	
	Ounces	Warp	Filling	Warp [1]	Filling	Warp	Filling
I	10.0	1	1	74	27	130	85
II	7.3	1	2	86	30	100	60

[1] Two yarns woven as one.

[2] A minus tolerance of 5 percent in breaking strength will be permitted in any one direction provided the combined strength equals the sum of the specified warp and filling requirements.

3.2.1 Weave. The weave shall be Oxford, 2 ends weaving as one (plain weave).

3.3 Width. The width, inclusive of selvage, shall be as specified (see 6.2). A tolerance of minus ¼ inch will be allowed on the specified width.

3.4 Nonfibrous materials. The starch and protein content, including chloroform-soluble and water-soluble material of the finished cloth shall be no more than 12.0 percent when tested as specified in 4.4.

3.5 Length and put-up. The cloth shall be put up in rolls in accordance with PPP–P–51. Unless otherwise specified (see 6.2), pieces shall be delivered in lengths of 80 yards or more. Pieces shall be sewn together with a butted FSa–1 type seam of Fed. Std. No. 751, using ticket No. 10, 3 ply unbleached cotton thread, conforming to type IIIA of V–T–276, and having 6 to 8 stitches per inch. The stitching shall be placed $\frac{1}{2} \pm \frac{1}{16}$ inch from the cut or raw edge of the cloth. The number of pieces in the roll shall be indicated by stencil marking on the last 6 inches of the roll on the outer surface.

3.6 Fiber identification. Each roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.7 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the

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specification where such inspections ade deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Inspection. Except when otherwise specified, inspection shall be in accordance with the provisions of MIL–STD–105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable procurement documents.

4.2.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of MIL–STD–105 and 4.2.2.1 through 4.2.3.

4.2.2.1 Yard-by-yard examination. The required yardage of each roll shall be examined on the face side and all defects as defined in Section I of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.2, except that only those slubs and knots which exceed the maximum acceptable limits shown for each in figure 1 shall be scored. No linear yard (increment of 1 yard on the measuring device) shall be penalized more than 4 points. The sample size shall be 750 yards, regardless of lot size. The sample yardage shall be examined from 15 rolls. An approximately equal number of yards shall be examined in each piece of the sample. The lot shall be unacceptable if the points per hundred square yards exceeds 40.00 points. Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Contracted width of cloth (inches)} \times 750} = \text{Points per 100 square yards}$$

4.2.2.2 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	—one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	—two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	—three points
For defects exceeding 9 inches in any dimension	—four points

NOTE: The following defects when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.
Objectionable odor.
Width—Less than minimum specified.
Excessive neppiness.

4.2.2.3 Examination for length.

4.2.2.3.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for put-up and length. Each defect shall be counted not more than once in each roll examined. The lot shall be unacceptable if two or more of the following defects are present:

DEFECTS

Any length found to be more than 2 yards less than the length marked on the roll ticket.

Length of piece:

(a) Any piece less than 80 yards in length.
(b) Pieces not sewn together with a butted FSa–1 seam.
(c) 6 to 8 stitches per inch not used in seam.
(d) Stitching not $\frac{1}{2} \pm \frac{1}{16}$ inch from the cut or raw edge of the cloth.

Number of pieces in roll not stenciled on last 6 inches of rolls on outer surface.

4.2.2.3.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths on the tickets.

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4.2.3 Examination for compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for compliance with the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.3 Examination of preparation for delivery requirements. Examination shall be made in accordance with the provisions of PPP–P–51 to determine that packaging, packing and marking requirements of section 5 of this specification are compiled with.

4.4 Testing of the end item. The methods of testing specified in CCC–T–191 wherever applicable and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit for testing shall be 1½ yards full width of the cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. All test reports shall contain the individual values utilized in expressing the final results. The sample size shall be as follows:

Lot size in yards	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE II Test methods

Characteristic	Requirement paragraph	Test method
Material identification	3.1.1	[1]
Yarn ply:		
Warp	3.1.1 and 3.2	Visual [2]
Filling	3.1.1 and 3.2	Visual [2]
Weight	3.2	5041
Breaking strength:		
Warp	3.2	5100
Filling	3.2	5100
Yarns per inch:		
Warp	3.2	5050
Filling	3.2	5050
Weave	3.2.1	Visual [2]
Nonfibrous materials	3.4	2611

[1] A certificate of compliance is required and will be acceptable for the requirements specified.
[2] One determination shall be made from each sample unit and the results reported as "pass" or "fail".

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2 and 6.2.1).

5.1.1 Levels A and C. The cloth put up as specified shall be packaged in accordance with the applicable requirements for class 4 fabrics of PPP–P–51.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2 and 6.2.1).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP–P–51.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP–P–51.

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6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for the following:

Type I is intended for use as insole reinforcements of various footwear items.

Type II is intended for use in the following specifications:

UU–B–346 —Binder, Loose-Leaf, Ring and Prong
UU–B–00355—Binder, Loose-Leaf, Prong
UU–B–00356—Binder, Loose-Leaf, Ring

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type and weight of cloth required (see 1.2 and 3.2).
(c) Width of cloth required (see 3.3).
(d) Length, if other than specified (see 3.5).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(f) Whether special marking is required (see 5.3).

6.2.1 In the preparation of contracts or orders it must be noted that various options, choices and alternatives as indicated in PPP–P–51 may be exercised in the preparation for delivery of cloth.

6.3 Supersession data. All weights listed in tables I and II of CCC–C–443b have been deleted except the 10 ounce in table I and the 7.26 ounce of table II.

Custodians:

Army—GL
Navy—SA

Review activities:

Army—GL, MD
Navy—SA

User activities:

Navy—OS, YD

Preparing activity:

Army—GL

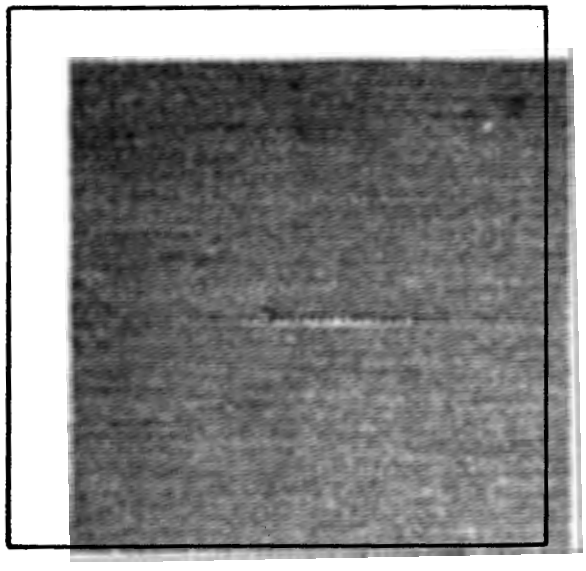
Civil Agencies Interest:

GSA-FSS
COM-NBS
JUS-FPI
VA

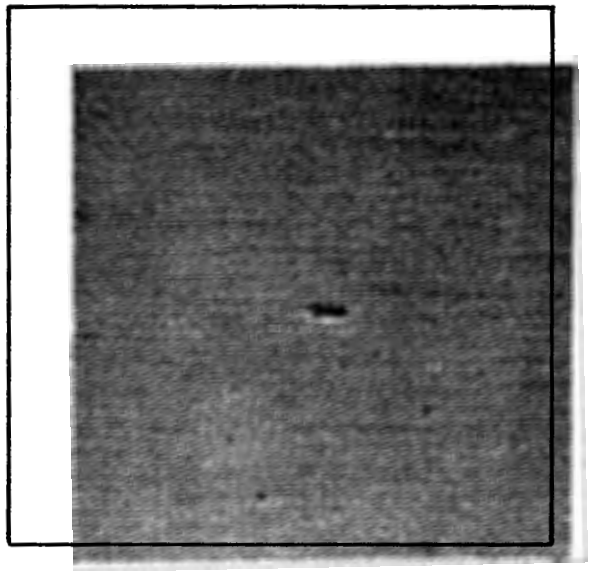
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Slub



Slub (Loose, Thick)



Lump or Knot (Fuzz)

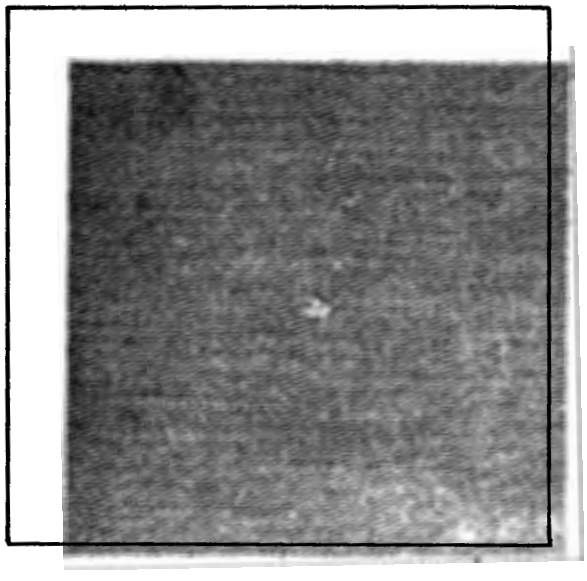


Figure 1 (Limits of Acceptability)

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION: Cloth, Duck, Cotton: (Single and Plied Filling Yarns, Flat) - CCC-C-443c

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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